

Software Requirements Specification

for

A System for Investor Profiling & Personalized Recommendations.

Version 1.0 approved

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Revision History

Name	Date	Reason For Changes	Version

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document aims to give a thorough overview of the functional and non-functional requirements for the Invest Map, an Investor Profiling and Recommendation System. The system provides individualized investment recommendations across worldwide capital markets to individuals and small to medium advising companies, taking global diversification, risk preferences, and ESG factors into account. The proposed system's design, implementation, testing, and validation are all built upon the SRS.

1.2 Document Conventions

This publication upholds better standards and quality by adhering to IEEE standards.

The entire document is written in Times New Roman typeface. The text body's font size is 12. The primary headings have a font size of 20. The subheadings have a font size of 16. The primary and subheadings are all bold. Floating point numbers are used to number each subsection.

1.3 Intended Audience and Reading Suggestions

The document is intended for the following audience:

Developers and Engineers: To understand the system's requirements and technical architecture for implementation.

Project Managers: To track milestones and ensure alignment with project goals.

Supervisors and Evaluators: To validate that the system meets academic and research objectives.

Testers: To design test cases based on specified requirements.

End Users and Advisors: To comprehend system functionalities and limitations.

1.4 Product Scope

Invest Map is a web based tool for investor profile and recommendation that is intended to offer individualized, transparent, and globally inclusive investment suggestions. Risk tolerance, investment horizon, industry focus, regional preferences, and Environmental, Social, and Governance (ESG) priorities are just a few of the preferences that investors can express through the system.

Using a curated dataset of about 400 investment instruments, including equities, commodities, indices, and exchange traded funds (ETFs), the software analyzes these inputs using a hybrid recommendation engine that combines rule based filtering with weighted scoring logic. An easy to use dashboard featuring ranking recommendations, comparative views, and exportable reports is used to display the results.

Purpose and Benefits:

Accessibility: Provides affordable advisory tools for small and medium sized advisory firms that lack resources for expensive proprietary systems.

Transparency: Ensures recommendation logic is explainable, building trust with investors and advisors.

Global Inclusion: Supports international diversification by including instruments from multiple regions and markets.

Responsible Investing: Integrates ESG considerations alongside traditional investment criteria.

Practical Demonstration: Demonstrates the feasibility of applying modern web technologies and hybrid recommendation approaches in financial advisory contexts.

This software contributes to the digital transformation of financial advisory services, particularly for resource constrained firms, advancing both academic understanding and practical applications in FinTech.

1.5 References

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2. Overall Description

2.1 Product Perspective

The suggested software is a brand new, standalone web based intelligent investment advising platform that aims to digitize and customize the process of investor profile and recommendation. It is an inventive system that combines several elements, such as rule based filtering, weighted scoring, ESG integration, and curated datasets into a single, cohesive solution designed for small and medium sized advisory firms and individual investors. It is not a continuation of any existing application.

Technology connects investors with tailored investment advice through a single platform. It provides precise, context aware services by integrating with several external data sources and technologies:

Investment Dataset: Curated collection of approximately 400 instruments (stocks, commodities, indices, ETFs) with attributes such as ticker, name, industry, region, currency, risk level, ESG score, volatility, and minimum investment.

Web Scrapping Tools: Collect and update financial instrument data from public sources including Yahoo Finance and Investing.com.

Frontend Framework: React.js for responsive, interactive user interfaces.

Backend Services: Python (Flask/FastAPI) for recommendation logic and API endpoints.

Database: MySQL for structured storage and efficient querying of investment data.

Future ML Integration: Architecture designed to accommodate machine learning models for enhanced personalization.

These components are orchestrated through a well defined API layer, which coordinates data retrieval, processing, and output generation for user interactions.

2.2 Product Functions

Invest Map integrates intelligent profiling, hybrid recommendation logic, and transparent reporting to enhance the investment decision making process. Its major functional modules include the following:

2.2.1 Investor Profiling Module

This module facilitates the collection of investor preferences through an intuitive web interface. Investors can specify:

- Risk tolerance (Low, Medium, High)
- Investment Horizon (Short, Medium, Long-term)
- Industry or sector preferences
- Geographic or regional focus
- ESG priority level (None, Moderate, High)
- Approximate investment capacity

The system validates and structures these inputs into a standardized investor profile for processing by the recommendation engine.

2.2.2 Investment Database

A curated dataset of approximately 400 investment instruments spanning multiple asset classes:

- Stocks: 200+ real global stocks from major exchanges (NYSE, NASDAQ, LSE, TSE, HKEX, NSE, etc.)
- Commodities: 50 real commodity instruments (gold, silver, oil, agricultural products, etc.)
- Indices: 60 real global indices (S&P 500, FTSE 100, Nikkei 225, etc.)
- ETFs: 90 real ETFs from providers like Vanguard, iShares, SPDR, and Invesco

Each instrument includes attributes such as ticker, name, industry, region, currency, risk level, volatility score, ESG score, market cap category, 1-year return (%), term suitability, and minimum investment.

2.2.3 Hybrid Recommendation Engine

The core intelligent component that processes investor profiles against the investment dataset using a two-stage approach:

- Rule-based filtering: Removes unsuitable investments based on constraints such as region, instrument type, and minimum investment.
- Weighted scoring: Calculates a match score for remaining investments using configurable weights for risk alignment, expected return, ESG score, and volatility.

The engine generates ranked recommendations with explanatory notes, ensuring transparency and explainability.

2.2.4 Dashboard and Reporting Interface

Presents recommendation results in a user-friendly format with:

- Ranked list of investment options with key attributes
- Visual indicators for risk level, ESG score, and match strength
- Comparison tools for side-by-side evaluation
- Exportable summary reports (PDF/Excel format)
- Broker connection information for interested investors
-

2.2.5 Admin Functions

Enables authorized users to:

- Update and expand the investment dataset
- Adjust scoring weights and profiling parameters
- Monitor system usage and performance
- Manage user access and roles

2.3 User Classes and Characteristics

Invest Map is designed to serve multiple categories of users, each with distinct roles, privileges, and interaction levels.

2.3.1 Investor

Individual investors seek personalized investment recommendations based on their preferences.

- Non-technical users require simple, intuitive interface
- Varying levels of financial literacy
- Expect privacy and data protection for personal information
- Access system via web browsers on desktop or mobile devices
- Rely on system for initial investment screening and comparison

2.3.2 Advisor

Professional financial advisors from small and medium-sized firms use the system to support client recommendations.

- Moderate technical literacy may require guided onboarding
- Need ability to interpret and explain recommendations to clients
- Value transparency and regulatory compliance
- Require reporting features for client communication
- May manage multiple client profiles

2.3.3 Admin

System administrators responsible for dataset management, user administration, and system configuration.

- Technically skilled staff with system-level privileges
- Manage investment dataset updates and quality assurance
- Configure scoring weights and profiling parameters
- Monitor system performance and user activity

2.3.4 System Integrator

Technical personnel responsible for system deployment, updates, monitoring, and performance optimization.

- High technical expertise in server configuration, database management, and API integration
- Infrequent direct interaction with user interface; focus on backend maintenance
- Responsible for ensuring system reliability and scalability

2.4 Operating Environment

Invest Map functions in a cutting edge, cloud based environment built for seamless integration, scalability, and cross platform accessibility. Users can easily access the system on platforms like Windows, macOS, Linux, Android, and iOS because it does not restrict usage depending on

operating systems. Since the system is web-based, installing any program is not necessary for its operation.

The main prerequisites for using the system are a reliable internet connection and a common web browser (such as Chrome, Firefox, Edge, or Safari). Consistent functionality and speed across devices and browsers are guaranteed by the system's cross platform usability optimization.

With MySQL database hosting, the backend is built to operate on cloud infrastructure (AWS, Azure, or comparable), enabling scalability as user demand increases.

2.4.1 Design and Implementation Constraints

The design and implementation of Invest Map are subject to several technical, regulatory, and operational constraints:

Constraint Type	Description
Technical Stack Constraints	Frontend must be developed using React.js and backend using Python (Flask/FastAPI)
Integration Constraints	System must interface with web scraping tools for data updates and support RESTful API architecture
Design and Coding Standards	Development must follow secure SDLC practices, modular coding conventions, and consistent API versioning
Storage and Data Constraints	System must efficiently handle the curated dataset (~400 records) within available cloud storage limits, with capacity for future expansion
Data Ethics Constraints	All numerical values (ESG, returns, volatility) must be synthetic for academic use.
Regulatory Constraints	System designed for advisory support only; no automated trading or execution features

2.5 User Documentation

Extensive external documentation is minimal because the system's architecture places a high focus on user friendliness. To give contextual aware instructions, the system strategically incorporates in app help features throughout the user interface. These include on screen alerts, tooltips, and succinct descriptions of key capabilities.

The user interface's simplicity and clarity help achieve this goal even more. The design's simple self explanatory layouts, iconography, and navigation routes make it easy for users to comprehend and interact with the system. Consistent linguistic and visual elements improve usability and lessen users' cognitive burden.

Additionally, the system adheres to best practices by utilizing succinct language, responsive design for device compatibility, and uniform formatting throughout. When combined, these elements guarantee a smooth and practical experience for each user.

2.6 Assumptions and Dependencies

Assumptions:

- Users have a device with minimum hardware requirements to access the system (smartphone, laptop, or desktop computer).
- Basic knowledge of technical device usage is expected.
- A stable internet connection is available.
- Investment datasets can be maintained and updated through publicly available sources.

Dependencies:

- Availability of public financial data sources (Yahoo Finance, Investing.com, IMF, World Bank) for dataset enrichment.
- Functionality of low code/no code platforms for rapid prototyping (if applicable).
- Cooperation of advisors for usability feedback during testing phases.
- University ethical clearance for research involving simulated user profiles

3. External Interface Requirements

3.1 User Interfaces

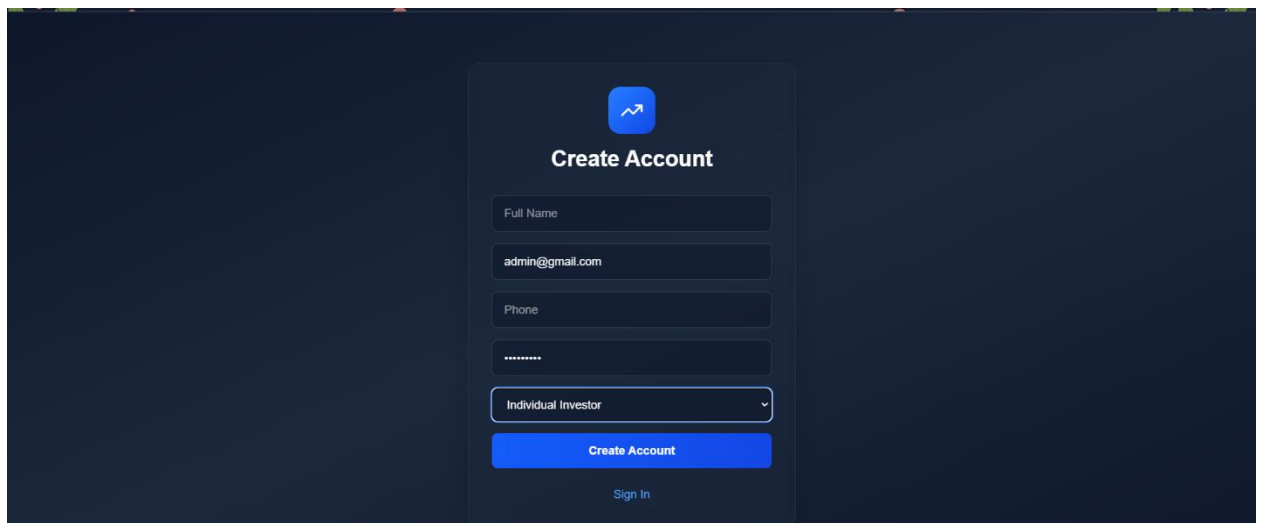


Figure 1: Create Account Page

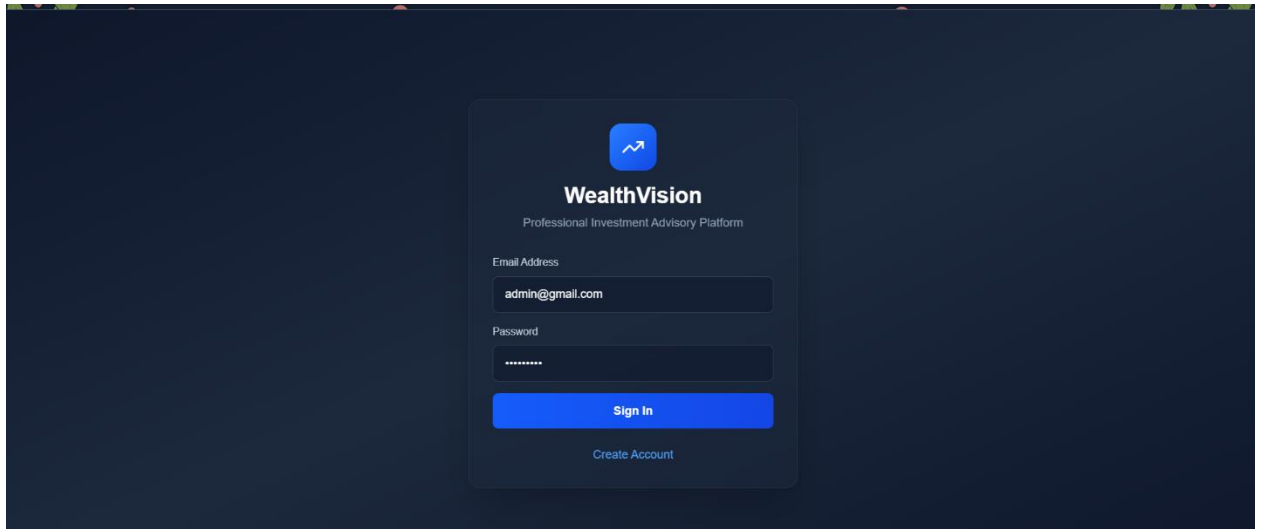


Figure 2: Login Page

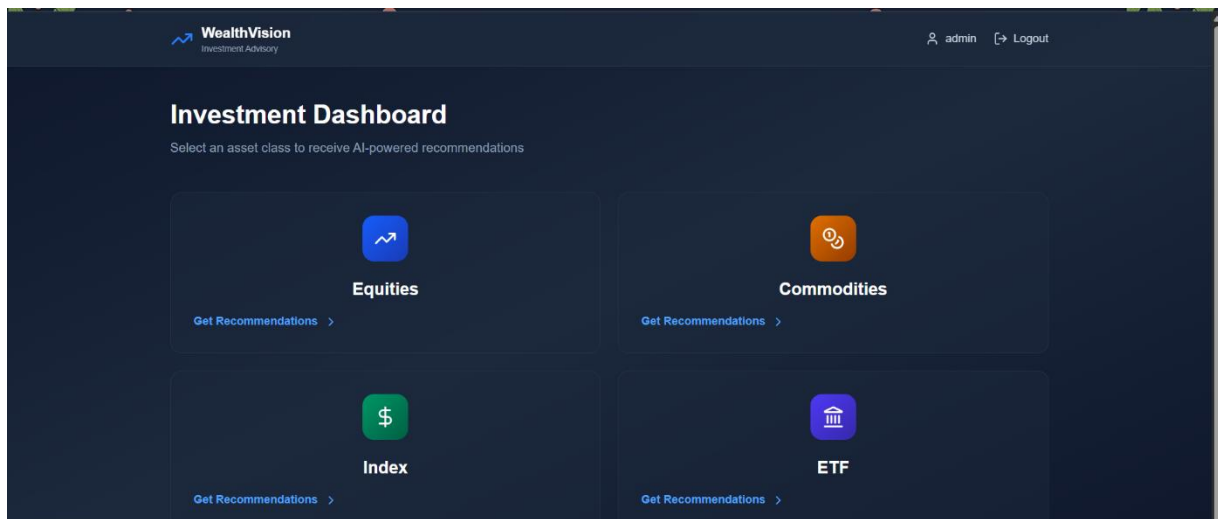


Figure 3: Dashboard

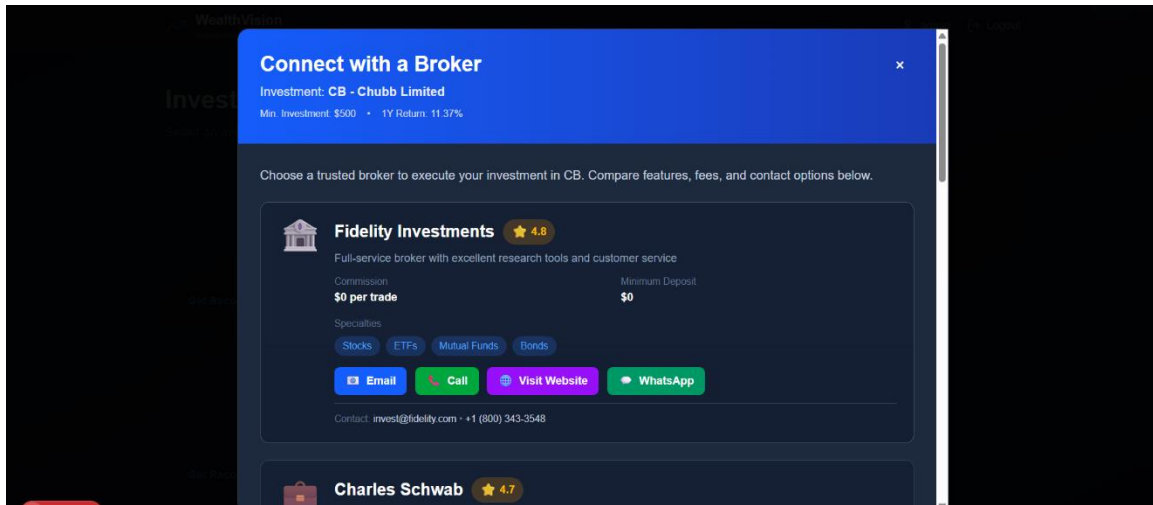


Figure 4: Broker Connect

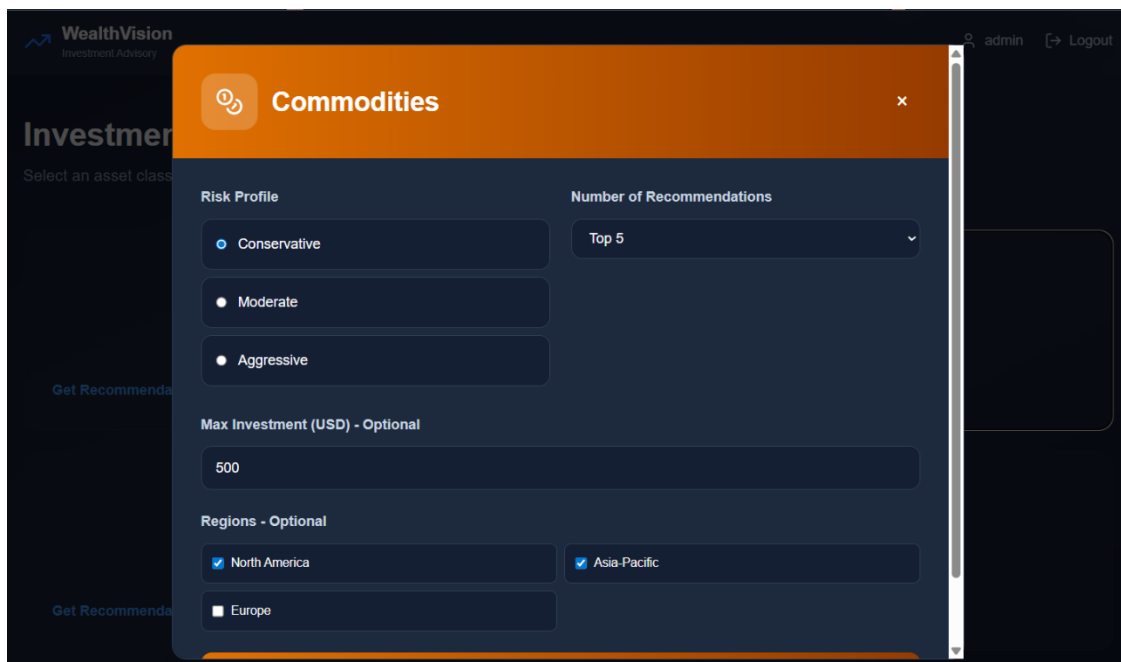


Figure 5: Selection Page

3.2 Hardware Interfaces

Desktops, laptops, and mobile devices are among the common hardware components with which the suggested system interacts. Compatibility with devices capable of running contemporary web browsers is one of the logical features. Interaction with input/output peripherals such as keyboards, mice, touchscreens, and network adapters is one of the physical features.

Hardware specific to the system is not needed. HTTP/HTTPS protocols control hardware and software communication, guaranteeing safe and effective data transfer.

3.3 Software Interfaces

Component	Technology	Purpose
Frontend Development	React.js, HTML5, CSS3, JavaScript	Interactive and responsive user interfaces
Backend Development	Python (Flask/FastAPI)	Server-side logic and RESTful APIs
Database	MySQL	Relational database for investment data and user profiles
Data Collection	Web scraping (Beautiful Soup, Scrapy)	Gathering instrument data from public financial sources
Recommendation Engine	Python with pandas, NumPy	Rule-based filtering and weighted scoring logic
Machine Learning (Future)	scikit-learn, TensorFlow	Predictive enhancements for personalization
Reporting	ReportLab, pandas	PDF and Excel report generation
API Architecture	RESTful APIs	Communication between frontend and backend

3.4 Communications Interfaces

Users connect to the database and backend services by using a web browser to access the application. Sensitive data is exchanged in an encrypted manner thanks to the HTTP/HTTPS protocols, which simplify communication. Form submissions, data retrieval, and dynamic content rendering are among the common web interactions supported by the interface.

Users and backend services can communicate seamlessly thanks to server client communication, which is based on RESTful API endpoints located on the backend server. API responses are formatted in JSON for efficient data exchange.

4. System Features

4.1 Investor Profiling and Recommendation Flow

4.1.1 Description and Priority

The entire process of gathering investor preferences, processing them via the hybrid recommendation engine, and providing tailored investment suggestions is covered by this feature. It is the Invest Map system's main feature.

Priority: High

Feature	Benefit	Penalty	Cost	Risk	Priority Score
Investor Profiling and Recommendation	9	7	5	4	7

$Priority\ Score = (Benefit + Penalty) - (Cost + Risk)$

4.1.2 Stimulus / Response Sequences

- Sequence 1: Investor Initiates Profiling

The investor goes to the profiling form after gaining access to the system.

Reaction: The profiling interface with all input fields and instructional text is shown by the system.

- Sequence 2: Investor Submits Preferences

The investor completes the form with risk tolerance, horizon, industry, region, ESG priority, and investment capacity.

Response: The system validates inputs, structures the data, and stores the investor profile for processing.

- Sequence 3: Rule-Based Filtering

The system applies rule-based filters to the investment dataset based on investor constraints.

Response: Investments that fail to meet minimum requirements (region mismatch, instrument type, budget constraints) are removed from consideration.

- Sequence 4: Weighted Scoring Calculation

The system calculates match scores for remaining investments using configurable weights.

Response: Each investment receives a composite score based on risk alignment, ESG match, expected return, and volatility.

- Sequence 5: Recommendation Generation

The system ranks investments by score and selects top recommendations.

Response: A ranked list of investment options with supporting attributes and explanations is prepared.

- Sequence 6: Results Presentation

The system displays recommendations on the dashboard.

Response: The investor views ranked recommendations with visual indicators, comparison options, and export functionality.

4.1.3 Functional Requirements

ID	Requirement
REQ-01	The system shall provide an interactive form capturing investor risk tolerance, investment horizon, industry preferences, geographic focus, ESG priority, and investment capacity.
REQ-02	The system shall validate all investor inputs and provide clear error messages for invalid entries.
REQ-03	The system shall maintain a curated dataset of approximately 400 investment instruments with comprehensive attributes.
REQ-04	The system shall apply rule-based filtering to remove investments that do not meet basic constraints.
REQ-05	The system shall calculate weighted scores for filtered investments using configurable weights.
REQ-06	The system shall generate ranked recommendations with explanatory notes for each suggestion.
REQ-07	The system shall display recommendations in a clear, sortable, and filterable dashboard interface.
REQ-08	The system shall allow users to compare multiple investment options side by side.
REQ-09	The system should provide exportable reports in PDF and Excel formats.

4.1.4 Investment Dataset Management

4.1.4.1 Description and Priority

This feature covers the creation, maintenance, and quality assurance of the investment dataset that powers the recommendation engine.

Priority: Medium

Feature	Benefit	Penalty	Cost	Risk	Priority Score
Investment Dataset Management	8	6	5	4	5

4.1.5 Stimulus / Response Sequences

- Sequence 1: Dataset Initialization

The system is deployed with an initial curated dataset.

Response: The dataset is loaded into the database with all required attributes.

- Sequence 2: Dataset Update

An administrator adds, modifies, or removes investment instruments.

Response: The system validates changes, updates the database, and maintains data integrity.

- Sequence 3: Data Quality Check

The system performs periodic validation of dataset attributes.

Response: Inconsistent or missing data is flagged for administrator review.

4.1.6 Functional Requirements

ID	Requirement
REQ-10	The system should maintain a structured database of investment instruments with consistent attributes.
REQ-11	The system shall support CRUD operations (Create, Read, Update, Delete) for investment data.
REQ-12	The system shall validate data integrity during updates to prevent corruption.
REQ-13	The system shall allow bulk import/export of dataset records via CSV files.
REQ-14	The system shall track dataset history for audit purposes.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

NFR Name	Description
Performance	• The system must maintain high responsiveness during concurrent user sessions. • Recommendation generation should complete within 3 seconds for typical queries. • The system should support at least 50 concurrent active users during peak load without service degradation. • Database queries must be optimized using indexing for efficient retrieval. • System uptime must meet a minimum of 99% availability, excluding scheduled maintenance periods. • Page load times should not exceed 2 seconds for standard operations.

5.2 Safety Requirements

NFR Name	Description
Safety	• The system must ensure data safety through regular automated backups. • Recommendations must include clear disclaimers that they are for informational purposes only. • The system must prevent any automated trading or execution functionality. • Emergency contact information for advisors/brokers should be clearly displayed. • Fallback mechanisms must be in place for API or data source failures. • The system should include safety notifications about investment risks.

5.3 Security Requirements

NFR Name	Description
Security	• The system must adhere to industry-standard security protocols to protect user privacy. • Communication between clients and servers must use HTTPS encryption. • User passwords must be hashed and salted using secure algorithms. • Authentication shall follow role-based access control (RBAC) for investors, advisors, and administrators. • Session management must include appropriate timeout and renewal mechanisms. • The system should undergo periodic vulnerability assessments. • No real personally identifiable financial information shall be stored without explicit consent.

5.4 Software Quality Attributes

Availability: The system should be available 24/7 with minimal planned downtime for maintenance.

Correctness: The system must provide accurate matching between investor profiles and investment options.

Flexibility: The system should allow addition of new features or dataset expansion without major restructuring.

Maintainability: Code should be modular, well-documented, and follow consistent conventions.

Reliability: The system should function without unexpected errors or crashes.

Testability: Components must be independently testable with clear, isolated responsibilities.

Usability: The interface must be intuitive, self-explanatory, and accessible to non-technical users.

Transparency: Recommendation logic must be explainable to build user trust.

Scalability: Architecture should support dataset growth and increased user load.

5.5 Business Rules

Function	Investor	Advisor	Admin	System
Register / Create Account	✓	✓	✓	
Complete Investor Profile	✓	✓		
Generate Investment Recommendations	✓	✓		
View / Compare Investments	✓	✓		
Export Recommendation Reports	✓	✓		
Manage Investment Dataset			✓	
Configure Scoring Weights			✓	
Manage User Accounts			✓	
View System Analytics			✓	✓
Perform Data Backups				✓
Monitor System Performance			✓	✓
Update System Configuration			✓	

6. Other Requirements

6.1 Database requirements

Database Scalability: The database must be built with flexibility to expand to thousands of instruments and manage growing volumes of investment data without compromising performance.

Data Backup and Recovery: To reduce downtime and possible data loss, automated daily backups and a recovery strategy must be put in place.

Data Integrity: All tables must preserve referential integrity.

Query Optimization: Frequently queried fields (instrument type, region, risk level, ESG score) require the creation of indexes.

6.2 Legal Requirements

Terms of Service and Privacy Policy: Both during registration and when using the system, the system must provide explicit terms of service and a privacy policy.

Compliance with Local Laws: The system must abide by all applicable local data protection laws.

Disclaimer Requirements: All investment recommendations must include a disclaimer indicating that they are not financial advice and are merely for informational purposes.

Ethical Research Compliance: When conducting research using simulated user data, the project must abide by the university's ethical requirements.

6.3 Data Requirements

Dataset Composition: Minimum 400 instruments across stocks (200+), commodities (50+), indices (60+), and ETFs (90+).

Data Freshness: To keep the dataset current, it should be checked and updated on a regular basis.

Attribute Completeness: All necessary characteristics must be included in every instrument; any missing values should be noted.

Synthetic Values: For academic purposes, ESG scores, returns, volatility, and risk levels must be artificially manufactured within reasonable ranges.

Appendix B: To Be Determined List

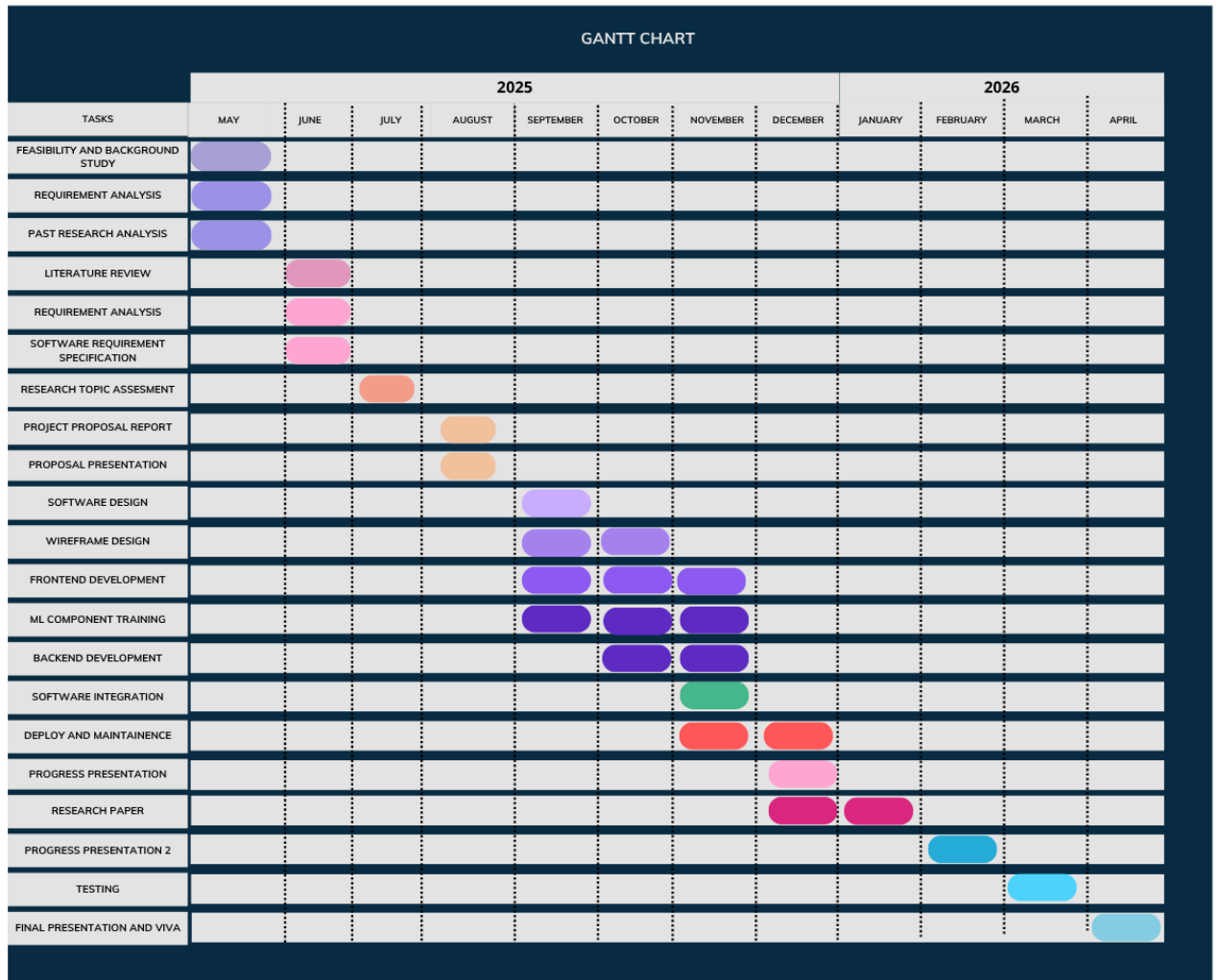


Figure 6: GanttChart